

AI exposure is rising in knowledge work while task automatability has not moved significantly: evidence from 12,487 job postings across five understudied industries, 2024 vs. 2026

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Public debate often fails to differentiate two separate questions about artificial intelligence and work: whether a job involves AI (exposure) and whether its tasks could be performed by AI (automatability). I separated them empirically. I scraped 12,487 LinkedIn postings from five understudied industries which includes farming/forestry, insurance, legal services, patent/IP, pharmaceutical/chemical at two snapshots, 2024 and 2026, and score each posting on both constructs with a cloud large language model (GPT) labeler, validated against an independent keyword measure and a 100-posting audit set. On a balanced, stratified sample of 2,614 postings, mean AI exposure rose from 0.061 to 0.098 between the two years ($p = 2.6 \times 10^{-8}$), an increase concentrated in knowledge-work fields (Patent/IP, Legal, Pharmaceutical/Chemical), while exposure dropped slightly in Farming and Insurance. Over the same period, mean automatability was statistically flat ($0.334 \rightarrow 0.324$, $p = 0.31$). At the posting level the two constructs do not have much of a correlation ($r = -0.11$, $n = 2,614$): a role's AI exposure says little about how automatable it is. A supervised baseline confirms the signal lives in description text (AUC 0.92) rather than structured metadata (AUC ≈ 0.69), motivating the language-model approach. Roles are increasingly built *around* AI without becoming measurably more *replaceable* by it.

artificial intelligence | labor demand | job postings | large language models | automation

Two claims about artificial intelligence and the labor market are often tied together, 1) that AI is becoming increasingly integrated into jobs, and 2) that AI is about to replace those jobs. These are different hypotheses. A machine-learning engineering role is saturated with AI yet is not a current candidate for automation, whereas a data-entry clerk's role may never mention AI yet consists almost entirely of codifiable, automatable tasks. I treat AI exposure and automatability as two separate measurement targets and ask how each changed between 2024 and 2026 across five industries that are underrepresented in existing studies.

Prior work measures AI's labor impact through occupation-level exposure indices and task-based automation scores (1–3). These frameworks disagree on what types of occupations or jobs is most exposed. Frey and Osborne found higher exposure in routine, lower-wage occupations (1), whereas Eloundou et al., studying large language models, find that higher-wage cognitive roles face the greatest exposure (2). Autor et al. show that augmentation and automation, though strongly correlated, carry opposite employment effects (4), while Acemoglu and Restrepo document causal job and wage losses from robot adoption (5) and the role of task automation in rising wage inequality (3). Yet this evidence concentrates on technology, finance, and office work, leaving more niche sectors such as agriculture, insurance, legal support, patent/IP services, and pharmaceutical/chemical manufacturing relatively understudied. I address both gaps by measuring exposure and automatability as two separate constructs, with a cloud language model, from employer-side demand signals (job postings), in five understudied industries.

Significance

Policy debates about artificial intelligence and jobs typically ask two different questions: whether work involves AI, and whether AI could do the work. Using 12,487 job postings from five understudied industries at two different snapshots two years apart, I measure these constructs separately with a cloud large-language-model labeler validated against an independent keyword measure and an audit set. AI exposure rose significantly between 2024 and 2026, concentrated in knowledge-work fields. On the other hand, the automatability of tasks did not change significantly, and the two measures are nearly independent. Treating “AI exposure” and “automation risk” as a single index can tell a misleading story on how AI is actually entering the labor market, as a tool built into professional work rather than a complete substitute for it.

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Parama Viprakasit designed the study, collected and analyzed the data, and wrote the paper.

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Table 1. Stratified analysis sample: postings per industry × snapshot (table08_sample_balance.csv).

Industry	2024	2026	Total
Farming, Ranching & Forestry	121	300	421
Insurance	300	300	600
Legal Services	300	300	600
Patent Analysis & IP Research	93	300	393
Pharmaceutical / Chemical Mfg.	300	300	600
Total	1,114	1,500	2,614

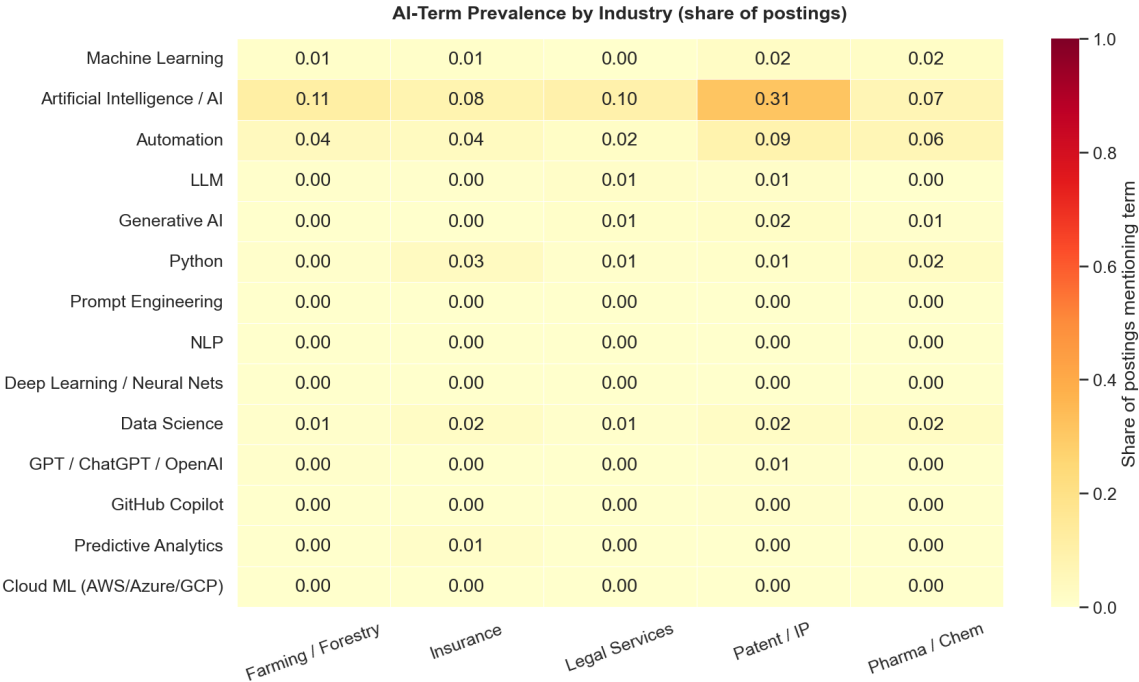


Fig. 1. Keyword AI-term prevalence by industry (notebook 03). This raw keyword signal is descriptive only: counts are inflated by 2026 descriptions running about 35% longer, which is why the analysis uses a length-robust language model rather than keyword counts.

Data

I collected job postings from LinkedIn for five industries at two points in two different years: an April 2024 job postings archive (source_platform = linkedin.2024) and a May 2026 scrape (source_platform = linkedin). After cleaning the data and removing duplicates (notebooks 01–02) the corpus contains 12,487 postings with title, full description, company, industry, seniority, and posting date. I also derived an independent keyword exposure measure: a binary flag set when a posting matches any of fourteen AI/ML-related terms (e.g., “machine learning,” “LLM,” “generative AI”). This serves as a transparent reference for validating the model-based score. Additional exploratory views (term-frequency distributions, the industry × AI-term heatmap of Fig. 1, and source-platform comparisons) are available in the project’s public website and repository.

A load-bearing caveat. Year is perfectly confounded with the scrape batch, so every 2024 posting comes from one source and every 2026 posting from another. This is therefore a two-snapshot comparison rather than a continuous time series, and I make no causal or temporal-trend claims. Because the industry mix also differs between scrapes, all comparisons are reported within industry. The keyword measure is further distorted by length. The 2026 descriptions run approximately 35% longer, inflating term counts compared to the 2024 set, so I rely on the length-robust language model score for all results. For scoring I draw a stratified sample of up to 300 postings per (industry × year) cell, totaling 2,614 postings (Table 1); two 2024 cells (Farming $n = 121$, Patent/IP $n = 93$) are thin and flagged throughout.

Methods

The labeler. Each posting is scored on two hypotheses, 1) *exposure*: jobs that involves artificial intelligence or machine learning; and *automatability*: jobs that consists mainly of routine tasks that a computer could automate. To choose the best model, I assembled a 100-posting audit set stratified across the five industries and labeled for both constructs, with five exposure and eighteen automatability positives that reflect the corpus’s low AI exposure/automatability positive rates, and benchmarked a panel of candidate labelers against it: a zero-shot BERT model (bart-large-mnli) and

Table 2. A supervised baseline predicting the keyword AI-exposure flag (model_comparison.csv): text features are necessary; metadata alone is near useless for the positive class.

Model (classification)	Test AUC	Test F_1
Logistic, metadata only	0.699	0.019
LightGBM, metadata only	0.687	0.000
Logistic + TF-IDF(2000)	0.901	0.612
LightGBM + TF-IDF(2000)	0.918	0.696
Ensemble + TF-IDF	0.917	0.695

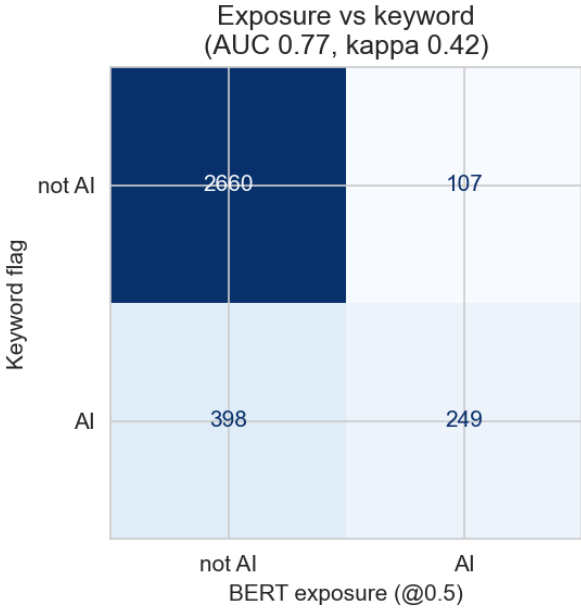


Fig. 2. Convergent validity of the exposure measure vs. the independent keyword flag (notebook 05): ROC-AUC 0.768, Cohen's $\kappa = 0.42$. Disagreements are mostly the model correctly withholding the label on boilerplate keyword hits.

several cloud language models from the GPT, Claude, and Gemini families (notebooks 06–07). My primary instrument is a **cloud GPT model** (gpt-5.4 for exposure, gpt-5.4-mini for automatability). BERT is retained as a free, reproducible cross-check and reproduces the same qualitative pattern, but noted that it significantly underperforms in the automatability metric.

Why a language model. A supervised baseline (notebook 04) shows that predictive signals of AI exposure and automatability live the job description text, not the structured metadata: predicting the keyword AI-exposure flag from metadata alone (industry, seniority, employment type) is weak (AUC ≈ 0.69 , $F_1 \approx 0.02$), while adding TF-IDF text features lifts it to AUC 0.92, F_1 0.70 (Table 2). An instrument that reads the text is therefore a higher-performing model, and a language model is able to read it without hand-built features.

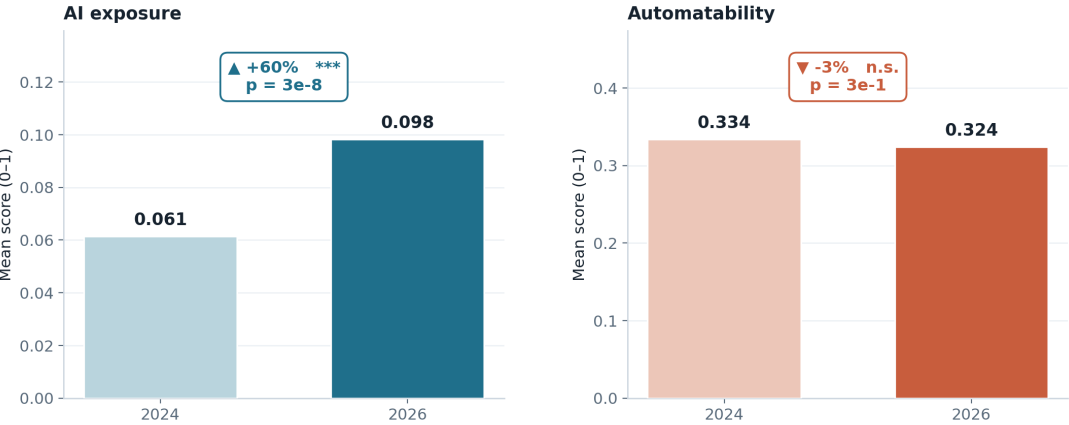
Validation. The two constructs are not equally validatable, so I validate each on its own terms. *Exposure* has an independent reference (the keyword flag): I report threshold-free convergent validity of ROC-AUC 0.768 against it (Fig. 2), rising to AUC 0.94 on the audit set, where the model beats the keyword baseline (F_1 0.57 vs. 0.42). *Automatability* has no ground-truth column, so I rely heavily on face validity; the top-scoring roles are clerical and coordination work while its lowest are senior, judgment-heavy roles; and discriminant validity against exposure. On the audit set the GPT automatability labeler reaches AUC 0.871 / F_1 0.61, whereas BERT's AUC indicates it is near-chance (AUC 0.60), the reason GPT, not BERT, is the headline labeler. I fix the decision threshold at 0.5. A sweep then confirms that exposure F_1 is flat across 0.4–0.6, so no test-set tuning is performed. The 2024 vs 2026 comparison uses continuous score means (t), which are independent from threshold. The audit labels are Claude-drafted and not independently human-reviewed, a limitation I return to below. Inspecting where the model and the keyword flag disagree is reassuring: the model most often *withholds* the AI label where the keyword fires on boilerplate (an agricultural-labor posting mentioning “innovation, technology” scores 0.02) and *assigns* it where AI work is described without the flagged terms (a blockchain-protocol engineering role scores 0.65 with no keyword match), reading the role itself, not the vocabulary.

Table 3. Main result, 2024 vs. 2026 ($n = 2,614$; table08_summary_by_year.csv). p -values are Welch's t on continuous GPT scores.

Measure	2024	2026	Δ	p
Mean exposure score	0.061	0.098	+0.037	2.6×10^{-8}
% exposed (≥ 0.5)	2.1	5.7	+3.6 pp	—
Mean automatability	0.334	0.324	−0.010	0.31
% automatable (≥ 0.5)	25.6	23.5	−2.1 pp	—

Roles increasingly involve AI — but aren't rated more automatable

2024 → 2026 · five understudied industries · the two constructs move in opposite ways



Model-labeled estimates · stratified sample $n=2,614$ (LinkedIn postings, 5 industries) · exposure: gpt-5.4 · automatability: gpt-5.4-mini

Fig. 3. Result: (notebook 08, GPT scores). Mean AI exposure rises significantly between snapshots while mean automatability does not move significantly.

Results

All results below use the GPT scores on the 2,614-posting stratified sample.

Exposure rose while automatability dropped slightly. Between the two snapshots, mean AI exposure increased from 0.061 to 0.098 ($t = 5.58$, $p = 2.6 \times 10^{-8}$) and the share of postings classified as AI-exposed increased from 2.1% to 5.7%. Over the same period mean automatability dropped slightly but was not statistically significant (0.334 → 0.324; $t = -1.01$, $p = 0.31$), and the automatable share decreased from 25.6% to 23.5% (Fig. 3, Table 3). The shift is not just a change in means, but also distributional, (Fig. 4): the 2026 exposure curve lies entirely to the right of the 2024 curve, while the two automatability curves are nearly overlapping.

The rise in exposure is concentrated in knowledge work. Within industry, exposure increased most in Patent/IP (+0.104), Legal Services (+0.068), and Pharmaceutical/Chemical (+0.045), while it dropped slightly in Farming (−0.021) and Insurance (−0.011) (Fig. 5). This concentration in cognitive, knowledge-heavy fields tracks the LLM-exposure pattern mentioned by Eloundou et al. (2) rather than the routine, lower-wage displacement pattern that Frey and Osborne found in their study (1). In contrast, automatability did not move much in any industry (all $|\Delta| \leq 0.04$). Figure 6 plots each industry's joint trajectory. Here, the clear direction is that industries slide rightward (exposure) but barely move vertically (automatability).

The two constructs are nearly independent. Across all 2,614 postings in the stratified sample, exposure and automatability barely correlate, with a correlation coefficient of only $r = -0.11$ ($p = 3.6 \times 10^{-8}$) which is statistically detectable but negligible in magnitude. Knowing a job/industry's AI exposure tells almost nothing about how automatable it is (Fig. 7). This is why the two constructs should not be collapsed into a single “AI risk” score. A posting can describe AI as a tool used inside a role without implying that the role's tasks are becoming more substitutable. In that sense, the result aligns with the augmentation–automation distinction from Autor et al. (4): AI can reshape work without necessarily replacing it.

The split is visible in the jobs themselves.. Splitting the sample at the 0.5 threshold makes the separation concrete (Table 4). Only 0.4% of postings are both AI-exposed and automatable. Another 3.7% are exposed but not automatable, while 24% are automatable but not exposed.

The two off-diagonal groups describe very different work. The exposed-but-not- automatable group contains roles such as Lead AI Research Scientist, Senior AI Engineer, AI Architect, and Data Scientist: jobs where AI is central to the work, but the work itself is not routine. The automatable-but-not-exposed group contains roles such as Scrubber Operator, Custodian, Production Line Associate, and Environmental Service Technician: jobs that are more routine, but rarely advertise themselves through AI language.

AI exposure rose between 2024 and 2026; automatability held steady

Cumulative share of postings by score — a curve further right means higher scores. Exposure shifts right in 2026; automatability barely moves.

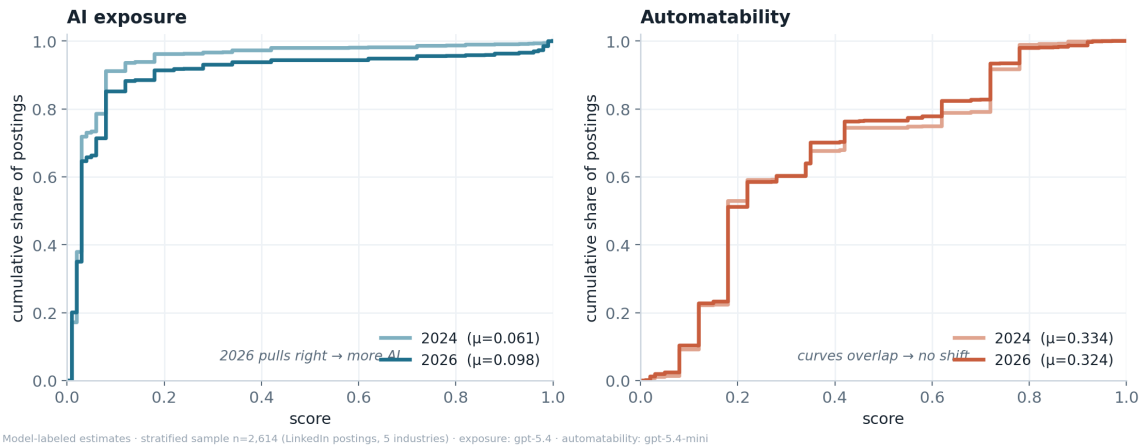


Fig. 4. Empirical CDFs of both scores by snapshot (notebook 08). The 2026 exposure curve sits entirely right of 2024 (consistently scoring higher throughout), and automatability’s two curves overlap in many spots. The shift is in the whole distribution rather than just the mean.

Table 4. Joint classification at the 0.5 threshold ($n = 2,614$). The two AI-positive cells barely overlap, and the off-diagonal cells hold opposite kinds of work.

	Automatable	Not automatable
AI-exposed	0.4%	3.7%
Not exposed	24.0%	71.9%

A single “AI risk” score would blur these cases together. Measuring exposure and automatability separately keeps the distinction visible: some jobs are being built around AI, while others are routine without being AI-facing.

Discussion

Across the five understudied industries that I selected for this study, employer demand became measurably more AI-involved between 2024 and 2026 without becoming more automatable. The results of this study addresses most directly to the question of which work is exposed which, based on this model, is heavily knowledge work, consistent with the LLM-exposure findings of Eloundou et al. (2) and the cross-country evidence of Georgieff and Hye (6), while the conclusions go against the routine-first view of Frey and Osborne (1). Exposure rose with no matching rise in automatability which aligns with Autor et al.’s distinction between augmentation and automation (4). While postings increasingly describe AI as a tool integrated into professional work, there is no strong evidence that it is currently a substitute for it. A possible mechanism is technological fit: large language models are strongest at language and document-centric tasks (drafting, search, synthesis, review) that makes up the core of patent, legal, and pharmaceutical-research work, so where the technology complements existing expertise employers advertise it, while in sectors where core tasks are physical (farming) or transactional (insurance), current models map less directly onto advertised roles and exposure does not rise. This is consistent with the Acemoglu and Restrepo’s task-based logic (3), where technology reshapes demand for the tasks it can perform. One caution may have affected the reading: because exposure is read from posting language, part of the knowledge-work rise may reflect faster adoption of AI vocabulary, which might be signaling AI-fluency to candidates rather than actual deeper integration, a distinction a text-based measure cannot fully identify or resolve. Because my measure is demand-side (reads advertised work, not headcounts) it does not reveal any implications or conclusions on net employment, the question on which recent evidence still diverges (rising unemployment in exposed occupations (7) versus within-firm reallocation (8)). It instead, reveals to how the content of advertised work is shifting. The flat automatability signal and the near-zero correlation between exposure and automatability together argue against combining the two into a single “AI risk” index. Two further checks point the same way. An unsupervised topic model (BERTopic; notebook 05) recovers the corpus’s themes without using any AI labels, and its most AI-exposed clusters (an AI and cloud-solutions theme concentrated in Patent/IP, together with cybersecurity, actuarial, and data-science topics) are the knowledge-work areas where exposure rose, while the largest, most routine clusters (clinical operations, general legal practice) carry relatively very low exposure. And a free zero-shot BERT model, scored on an independent sample, reproduces the same within-industry exposure rise and flat automatability, so the headline is not an artifact of any single labeler. Cross-country and OECD analyses, similarly, find AI’s aggregate employment footprint to be relatively small in the big picture, but unevenly distributed across skills and wages (6, 9, 10).

AI exposure diverges sharply by industry, 2024 → 2026

Knowledge work surges (Patent/IP, Legal, Pharma); manual & clerical roles hold flat or dip

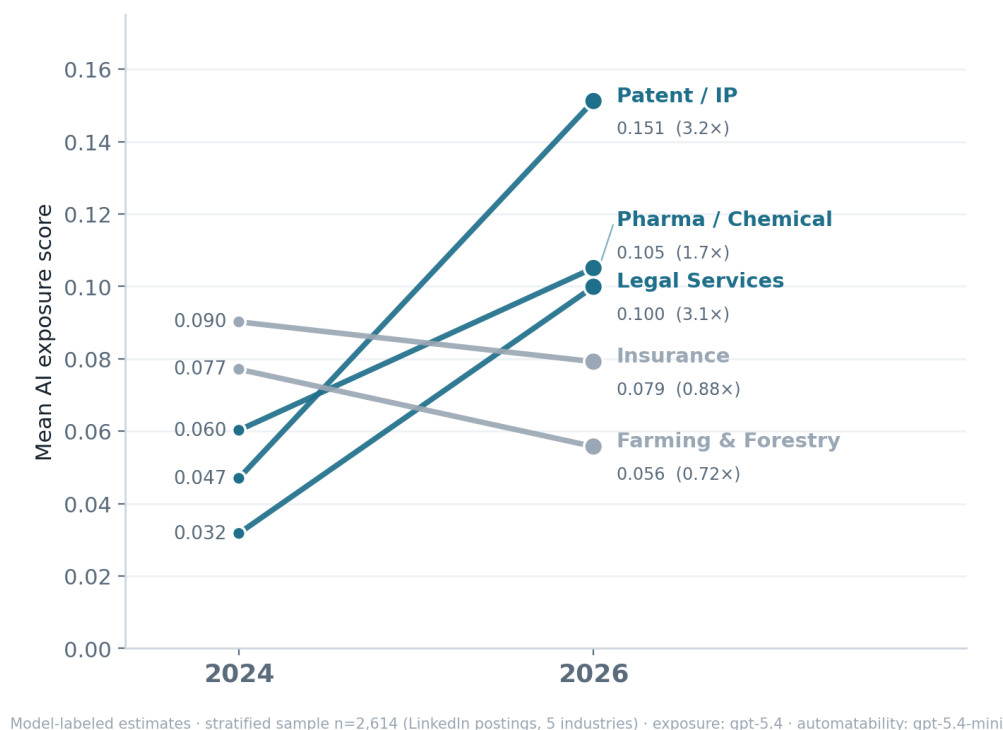


Fig. 5. Mean AI exposure by industry, 2024 to 2026 (notebook 08). Knowledge-work fields (Patent/IP, Legal, Pharma) surge; manual and clerical fields hold flat or dip.

Limitations. (i) *Design:* year is confounded with scrape source, so this study is a two-snapshot contrast between two separate years, not a causal or temporal trend; all claims are within-industry, and two 2024 cells are thin. (ii) *Measurement:* the automatability construct lacks an external ground-truth benchmark. I audited a 100-posting validation set, but the labels were AI-drafted and the audit included only five exposure-positive cases, so the absolute automatability estimates should be treated as directional. (iii) *Sampling:* LinkedIn over-represents formal, English-language, white-collar postings. (iv) *Construct:* “automatable” is inferred from description text, which may understate routine manual tasks that postings do not spell out. Future work should expand the audit set with more human-verified positives, add intermediate time points, and triangulate postings against task-level data.

Agentic Analysis

I used an AI coding assistant (Claude-Opus-4.8) throughout the pipeline and fed it ideas on what types of analysis and models to test: scaffolding the scraping and preprocessing notebooks, suggesting and implementing designs for the two-construct measurement, selecting and validating the labeler, and improving figures and tables.

What I accepted. The core two-construct framing (exposure vs. automatability) as a way to address the Frey-Osborne and Eloundou disagreement; a validity-first workflow where the results were not written up or interpreted until the measure had been validated against the keyword reference and the audit set; and a posting-level density plot as a statistically honest method of displaying the exposure-automatability correlation.

What I rejected or corrected. The assistant’s first automatability hypothesis (“... could be performed by artificial intelligence”) initially collapsed onto exposure ($r = 0.83$) with inverted face validity, causing AI-development roles to score as most replaceable. To address this, I reworded it toward routine-task automatability, which separated the constructs ($r = -0.11$). I also rejected a two-construct figure that invited reading a correlation off only five industry-mean points and replaced it with the $n = 2,614$ posting-level density after the small- n issue was identified.

Where it failed. The assistant repeatedly surfaced stale numbers (a superseded topic count and pre-rerun means) that I caught only by re-checking against the actual outputs; it once produced a topic model with no fixed random seed, so the topic count changed between runs which was a reproducibility gap I had to flag and control. The takeaway is similar to my experience working with NotebookLM, where these tools perform well on focused, verifiable subtasks, but weak when it came to unattended synthesis, and every quantitative claim must be checked against its source before being trusted.

The two-construct story in one map

Industries slide sideways (AI exposure shifts) but barely move vertically (automatability ≈ flat)

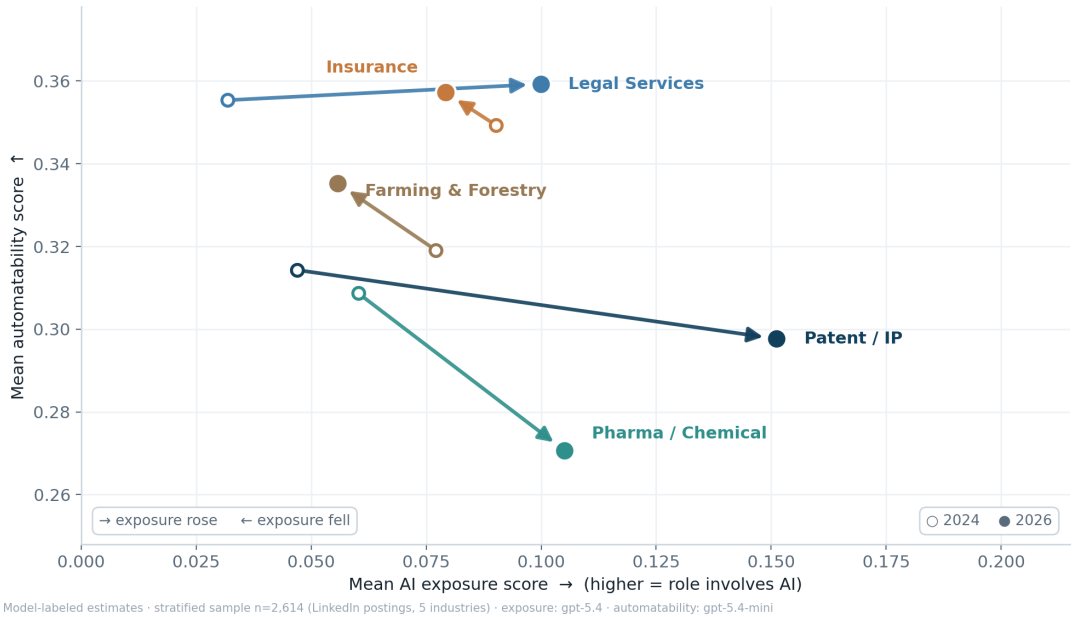


Fig. 6. Each arrow is one industry, 2024 to 2026, in exposure × automatability space (notebook 08). Large horizontal shifts, negligible vertical ones. This five-point view is descriptive; the correlation below is computed at the posting level.

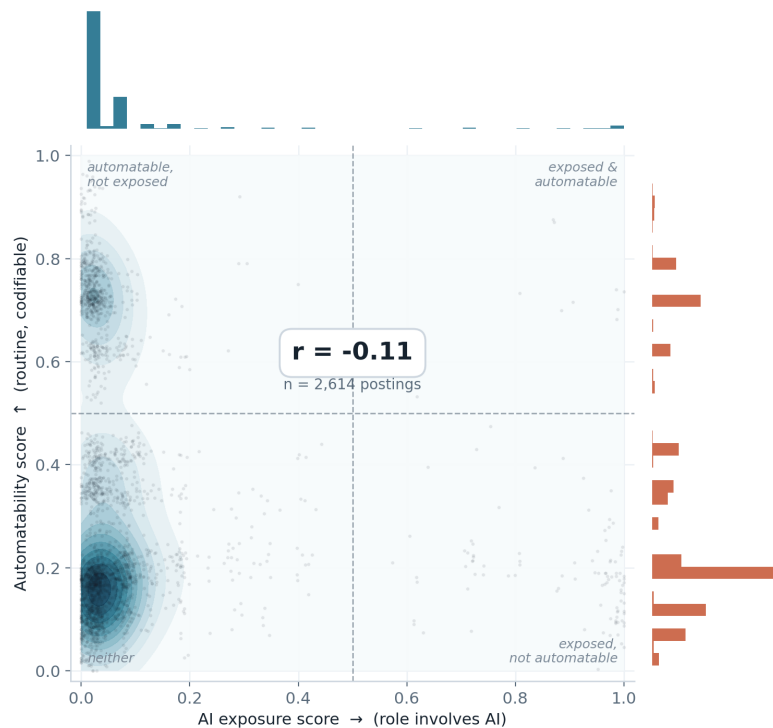
Data, Materials, and Software Availability. All code (numbered notebooks 00–09), data, and outputs, including the additional exploratory, supervised, and topic-model figures referenced above, are available at <https://github.com/peakviprakasit/QSS45--AI-Exposure-vs.-Automatability-in-Five-Understudied-Industries>. The project website is available at <https://parama-viprakasit-qss-projects.vercel.app/>. The cleaned corpus, model scores, and validation-audit set are under the repository’s data/ and output/ directories

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Exposure and automatability are nearly independent

Each point is one job posting ($n=2,614$), scored on both axes; AI exposure barely predicts how automatable a role is ($r = -0.11$).



Model-labeled estimates · stratified sample $n=2,614$ (LinkedIn postings, 5 industries) · exposure: gpt-5.4 · automatability: gpt-5.4-mini · scores jittered for legibility

Fig. 7. Posting-level density of the two GPT constructs ($n = 2,614$; notebook 08). AI exposure barely predicts how automatable a role is ($r = -0.11$). Marginal histograms show exposure concentrated near zero with a growing high-exposure tail, automatability is more broadly spread.